FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

JEE (Advanced)-2020

PAPER – 1

Time Allotted: 3 Hours **Maximum Marks: 186**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

1. ***Section–A (01 – 04, 19 – 22, 37 – 40)*** *contains 12 multiple choice questions which out of 4 options have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (05 – 12, 23 – 30, 41 – 48) *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*

   Full Marks : ***+4*** *If only (all) the four option(s) is (are) chosen.*
   Partial Marks : ***+3*** *If all the four options are correct but ONLY three options are chosen.*
   Partial Marks : ***+2*** *If three or more options are correct but ONLY two options are chosen and both of which are correct options.*
   Partial Marks : ***+1*** *If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   Zero Marks : ***0*** *If none of the options is chosen (i.e. the question is unanswered).*

   Negative Marks : ***–1*** *In all other cases.*

2. ***Section-C (13 – 18, 31 – 36, 49 – 54)*** *contains 18 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

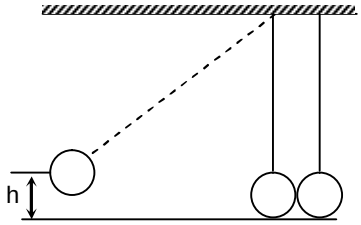
PART – I

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

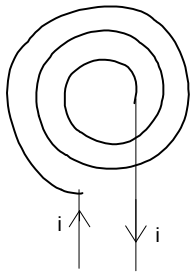
1. Two glass plates are separated by water. If surface tension of water is 75 dynes per cm and area of each plate wetted by water is 8 cm^2 and the distance between the plates is 0.12 mm, then the force applied to separate the two plates is
(A) 10^2 dynes (B) 10^4 dynes
(C) 10^5 dynes (D) 10^6 dynes

2. In the arrangement shown, the pendulum on the left is pulled aside. It is then released and allowed to collide with other pendulum which is at rest. A perfectly inelastic collision occurs and the system rises to a height $\frac{1}{4}$h. The ratio of the masses of the pendulums is

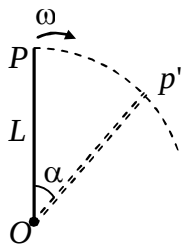


(A) 1 (B) 2
(C) 3 (D) 4

3. A thin insulate wire forms a plane spiral of N = 100 turns carrying a current I = 2.3 A. The inner and outer radii are equal to a = 5 cm and b = 10cm. Find the magnetic field (In 10^{-3} T) at the centre of spiral [In 2 = 0.69] is
(A) 1 (B) 2
(C) 3 (D) 4



4. A rod of uniform mass and of length L can freely rotate in a vertical plane about an axis passing through O. The angular velocity of the rod when it falls from position P and P' through and angle α is
(A) $\sqrt{\frac{6g}{5L}}\sin\alpha$ (B) $\sqrt{\frac{6g}{L}}\sin\frac{\alpha}{2}$
(C) $\sqrt{\frac{6g}{L}}\cos\frac{\alpha}{2}$ (D) $\sqrt{\frac{6g}{L}}\sin\alpha$

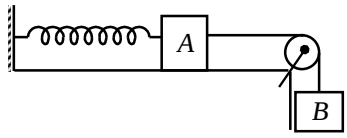


Space for Rough work

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

5. In the adjoining figure block A is of mass m and block B is of mass 2m. The spring has a force constant k. All the surfaces are smooth and the system is released from rest with spring unstretched



(A) The maximum extension of the spring is $\frac{4mg}{k}$

(B) The speed of block A when extension in spring is $\frac{2mg}{k}$, is $2g\sqrt{\frac{2m}{3k}}$

(C) Net acceleration of block B when the extension in the spring is maximum, is $\frac{2}{3}g$.

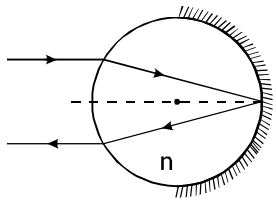
(D) Tension in the thread for extension of $\frac{2mg}{k}$ in spring is mg.

6. 1 gm of a radioactive substance takes. 50 sec to loose 0.01 gm then the half life of the sample will be

(A) $\frac{50\ln(2)}{\ln\left(\frac{100}{99}\right)}$ (B) $\frac{50\ln(2)}{\ln(100)}$

(C) $\frac{50\ln(2)}{\ln(99)}$ (D) $\frac{50\ln(2)}{\ln(0.99)}$

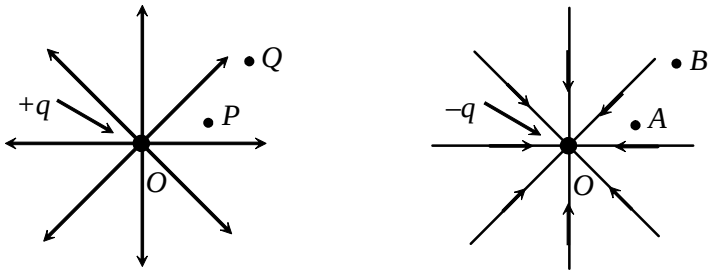
7. A transparent cylinder has its right half polished so as to act as a mirror. A paraxial light ray is incident from left that is parallel to principal axis, exits parallel to the incident ray as shown. Find the refractive index n of the material of the cylinder.



(A) 2 (B) 4
(C) 8 (D) 0

Space for Rough work

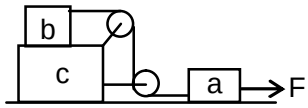
8. Figures shows the lines of force of the electric field of a positive charge (+q) and a negative charge (–q) respectively. Which of the following statements are correct?



(a) (b)

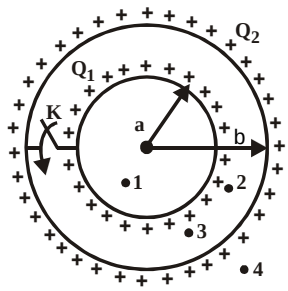
(A) potential at P is greater than that at Q.
(B) potential at A is greater than that at B.
(C) potential at P is less than that at Q.
(D) potential at B is greater than that at A.

9. Three blocks a, b, and c of masses 10 kg, 10 kg and 20 kg are arranged as shown in figure. All the surfaces are frictionless and string is inextensible. Pulleys are light. A constant force F = 20 N is applied on block a as shown. Pulleys and string are light. Part of the string connecting both pulleys is vertical and part of the strings connecting pulleys with blocks a and b are horizontal.



(A) Acceleration of mass blocks a, b and c is 0.5 m/s^2.
(B) Acceleration of mass block b is 1.
(C) Tension in the string is 10 N.
(D) Acceleration of mass block c is 0.5

10. Two concentric conducting shells as shown in figure having charge Q_1 and Q_2. If switch K is closed find points where potential will be same



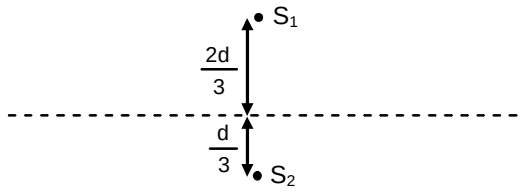
(A) 1 (B) 2
(C) 3 (D) 4

Space for Rough work

11. A charge q is moving with a velocity $\vec{V}_1 = 1\hat{i}$ m/s at a point in a magnetic field and experience a force $\vec{F}_1 = q[-1\hat{j} + 1\hat{k}]N$. If the charge is moving with a velocity $\vec{V}_2 = 1\hat{j}$ m/s at the same point, it experience a force $\vec{F}_2 = q[1\hat{i} - 1\hat{k}]N$. If the magnetic induction $\vec{B}$ at that point is $\vec{B} = a\hat{i} + b\hat{j} + c\hat{k}$ calculate a + b + c.

(A) 1 (B) 2
(C) 3 (D) 4

12. Two coherent point sources of frequency ($f = \frac{10v}{d}$ where v is speed of light in air) are placed at a distance d apart as shown in figure. The receiver is free to move along the dotted line shown in the figure. The total number of maxima observed by receiver is

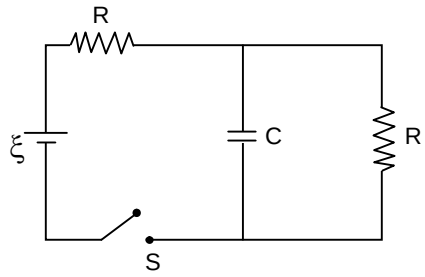


(A) 2 (B) 4
(C) 6 (D) 8

SECTION – C
(Numerical Answer Type)

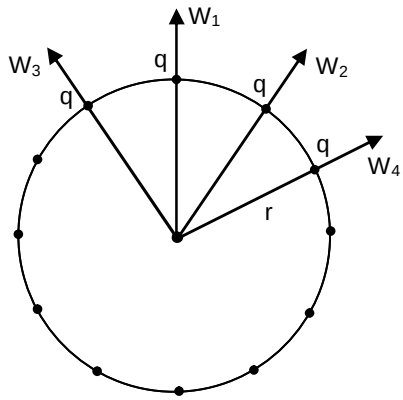
This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. XXXXX.XX).

13. As situation shown in figure if the maximum value of rate of energy stored in the capacitor after the switch is closed is $n\frac{\xi^2}{R}$ then find the value of 'n'.



Space for Rough work

14. N identical charges each of charge 'q' are symmetrically placed on the circumference of circle of radius 'r'. Four adjacent charges are slowly taken to infinity in the sequence a shown in figure. If W_1, W_2, W_3 and W_4 is the respective work done in slowly displacing q_1, q_2, q_3 and q_4. If $(W_3 - W_4)$ is $x\left[\frac{-q^2}{4\pi t_0 r \sin\frac{3\pi}{n}}\right]$ then

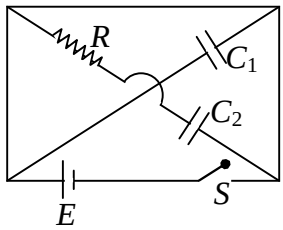


15. A particle is projected with speed u at an angle θ (< 60°) with the horizontal. If the time taken by it, so that its velocity makes an angle 60° with the initial velocity is $n\frac{y}{g}\frac{1}{\sin(\theta+30^o)}$

16. The equation of a particle executing SHM is given by $x = 3\cos\left(\frac{\pi}{2}\right)t$ cm, where *t* is in second. Find the distance travelled by the particle in the first 8.5 s

17. Three infinitely long thin conductors are joined at the origin of coordinates and lies along the *x*, *y* and *z* axes. A current *i* flowing along the conductor lying along the negative *x*-axis divides equally into the other two at the origin. If the magnitude of magnetic field at the point (0, –*a*, 0) is $n\frac{\mu_0 i}{4\pi a}$ then find the value of 'n'.

18. In the circuit shown, the capacitors C_1 and C_2 have capacitance *C* each. The switch *S* is closed at time $t = 0$. Taking $Q_0 = CE$ and $\tau = RC$, if the charge on C_2 after time $t = RC\, n\theta_0$ then find the value of 'n'.



Space for Rough work

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

19. Which of the following is not a property of NaOH?
(A) It is hygroscopic in nature
(B) It decolourizes chlorine gas
(C) It behaves as a dehydrating agent
(D) Heat is liberated when it reacts with HCl

20. For the gaseous reaction, $A(g) \rightarrow 4B(g) + 3C(g)$ is found to be first order with respect to A. If at the starting the total pressure was 100 mm Hg and after 20 minutes it is found to be 400 mm Hg. The rate constant of the reaction is
(A) 20 min^{-1} (B) 1.2×10^{-3} sec^{-1}
(C) 5.7×10^{-4} sec^{-1} (D) 3.4×10^{-2} sec^{-1}

21. Which of the following compound is formed if PbS reacts with H_2O_2?
(A) $PbSO_3$ (B) $PbSO_4$
(C) PbS_2O_3 (D) PbS_4O_6

22. $Zn(s)|Zn^{2+}(aq)||Cu^{2+}(aq)|Cu(s)$
Which of the following activity can increase the cell potential of the above electrochemical cell?
(A) Addition of water to the anode half-cell
(B) Addition of water to the cathode half-cell
(C) Increasing temperature
(D) Addition of same amount of water to both half-cells

Space for Rough work

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

23. In which of the following option(s) the dipole moment of the left side molecule is greater than that of the right side molecule?
(A) NH_3, NF_3 (B) SCl_2, $BeCl_2$
(C) SF_4, CF_4 (D) PF_5, IF_5

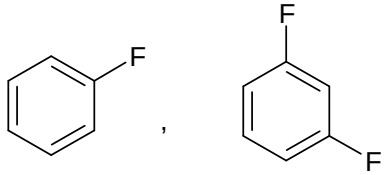
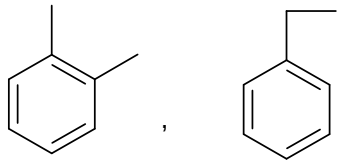
24. K_C for the reaction 2A (g) + B(g) $\rightleftharpoons$ A_2B(g) is 2×10^{-2} at 25°C and it is 2×10^{-1} at 50°C. Choose the incorrect statement (s) from the following
(A) Decomposition of A_2B is favoured at low temperature
(B) Formation of A_2B is favoured at low pressure.
(C) Formation of A_2B is favoured by introducing an inert gas at constant pressure.
(D) Decomposition of A_2B is favoured by removing B(g) from the reaction mixture

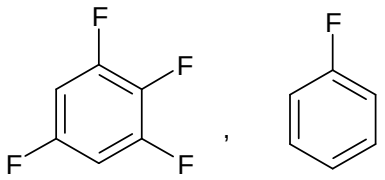
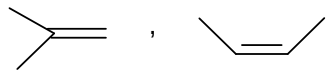
25. Which of the following ion(s) form(s) precipitate with NaOH solution?
(A) Ca^{2+} (B) Fe^{3+}
(C) Cr^{3+} (D) Al^{3+}

26. Which of the following ion(s) maintain an identical d-orbital configuration while forming octahedral complexes with strong field as well as weak field ligands according to crystal field theory?
(A) Fe^{2+} (B) Co^{2+}
(C) Ni^{2+} (D) Zn^{2+}

Space for Rough work

27. Which of the following pair has same net dipole moment

(A)  (B) 

(C)  (D) 

28. Which of the following are the common metallurgical processes which are used for the extraction of zinc from ZnS and $ZnCO_3$?

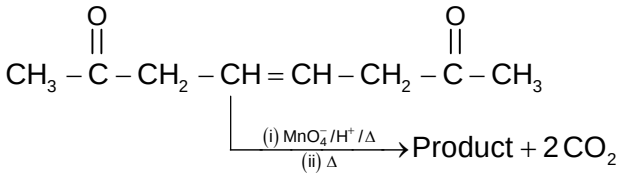
(A) Calcination (B) Roasting

(C) Carbon reduction (D) Distillation

29. In NaCl unit cell Na^+ ions are present at

(A) corners of the unit cell (B) face centres of the unit cell

(C) edge centres of the unit cell (D) body centre of the unit cell

30. 

How many carbon atom(s) is/are present in the unknown product of above reaction?

(A) 3 (B) 4

(C) 1 (D) 6

Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. XXXXX.XX).

31. What is the sum of azimuthal quantum number of all the valence electrons of aluminum? [At. No. of Al = 13]

32. How many S – O single bonds are present in Caro's acid (H_2SO_5)?

33. The formula of a spinel is $MgAl_2O_4$. How many effective no. of anionic vacancy for unit cell will be created if the oxide ions of the spinel are replaced by $X^{-8/3}$ ions?

34. The radiator of a car contains 52 g water. If 6.2 g of glycol ($HOCH_2CH_2OH$) is added to it, the solution freezes at –4°C. How many gram of ice will be separated in the process? [K_f = 1.86 K kg mol^{-1}]

35. What is the pH of a solution which contains CH_3COOH and it's sodium salt in the molar ratio of 10 : 1?
[pK_a of CH_3COOH = 4.74]

36. The r.m.s velocity of NO gas at 500 K is equal to the most probable velocity of an unknown gas at 314 K. What is the molecular mass of the unknown gas?

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. The locus of mid points of the chords of the circle $x^2 - 2x + y^2 - 2y + 1 = 0$ which are of unit length is

(A) $(x-1)^2 + (y-1)^2 = \frac{3}{4}$ (B) $(x-1)^2 + (y-1)^2 = 2$

(C) $(x-1)^2 + (y-1)^2 = 4$ (D) none of these

38. Suppose we define the distance between two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ as $d(P, Q) = \max.\{|x_2 - x_1|, |y_2 - y_1|\}$, then the area of the region bounded by the locus of a point P satisfying $d(P, A) = 4$, where A is (1, 2) is

(A) 64 sq. units (B) 54 sq. units

(C) 16π sq. units (D) none of these

39. Let $z_1 = a + ib$ and $z_2 = c + id$ be two complex numbers such that $|z_1| = |z_2| = 1$ and $\mathrm{Re}(z_1\bar{z}_2) = 0$. If a, b > 0 and c < 0, then

(A) $0 \le |z_1 - \bar{z}_2| \le 2$ (B) $0 < |z_1 - \bar{z}_2| < \sqrt{2}$

(C) $\sqrt{2} < |z_1 - \bar{z}_2| \le 2$ (D) $\sqrt{2} < |z_1 - \bar{z}_2| < 2$

40. If $a_1, a_2, a_3 (a_1 > 0)$ are in G. P. with common ratio r, then the value of r, for which the inequality $9a_1 + 5a_3 \le 14a_2$ holds, can not lie in the interval

(A) $[1, \infty)$ (B) $\left[1, \frac{9}{5}\right]$

(C) $\left[\frac{4}{5}, 1\right]$ (D) $\left[\frac{5}{9}, 1\right]$

Space for Rough work

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

41. If $\int \frac{xe^x}{\sqrt{(1+e^x)}}dx = f(x)\sqrt{(1+e^x)} - 2\ln g(x) + c$, then

(A) $f(x) = x - 1$

(B) $g(x) = \frac{\sqrt{(1+e^x)}-1}{\sqrt{(1+e^x)}+1}$

(C) $g(x) = \frac{\sqrt{(1+e^x)}+1}{\sqrt{(1+e^x)}-1}$

(D) $f(x) = 2(x-2)$

42. Which of the following are true:

(A) $\sqrt{e^\pi} > \left(\frac{\pi}{2}\right)^e$

(B) $e^\pi > \pi^e$

(C) $e^3 > 3^e$

(D) $e^2 > 2^e$

43. If ${}^{100}C_6 + 4.{}^{100}C_7 + 6.{}^{100}C_8 + 4.{}^{100}C_9 + {}^{100}C_{10}$ has the value equal to ${}^xC_y$; then the possible value (s) of $x + y$ can be:

(A) 112
(B) 114
(C) 196
(D) 198

44. If $\vec{a} = \hat{i} + 6\hat{j} + 3\hat{k}$; $\vec{b} = 3\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{c} = (\alpha+1)\hat{i} + (\beta-1)\hat{j} + \hat{k}$ are linearly dependent vectors and $|\vec{c}| = \sqrt{6}$; then the possible value(s) of $(\alpha+\beta)$ can be:

(A) 1
(B) 2
(C) 3
(D) 4

Space for Rough work

45. If $A=\begin{bmatrix}3 & -3 & 4\\ 2 & -3 & 4\\ 0 & -1 & 1\end{bmatrix}$, then

(A) $\text{adj}(\text{adj}\,A)=A$ (B) $\left|\text{adj}(\text{adj}\,A)\right|=1$

(C) $\left|\text{adj}\,A\right|=1$ (D) none of these

46. There exists a triangle ABC satisfying the conditions

(A) $b\sin A=a,\ A<\frac{\pi}{2}$ (B) $b\sin A>a,\ A>\frac{\pi}{2}$

(C) $b\sin A>a,\ A<\frac{\pi}{2}$ (D) $b\sin A<a,\ A<\frac{\pi}{2},\ b>a$

47. If $I_n=\int_0^{\pi/4}\tan^n x\,dx,$ then

(A) $I_5+I_3=\frac{1}{4}$ (B) $I_8+I_6=\frac{1}{7}$

(C) $I_6-I_{10}=\frac{2}{63}$ (D) $I_6+2I_8+I_{10}=\frac{16}{63}$

48. If A, A_1, A_2, A_3 are the areas of the inscribed and escribed circle of a $\Delta ABC,$ then

(A) $\sqrt{A_1}+\sqrt{A_2}+\sqrt{A_3}=\sqrt{\pi}\left(r_1+r_2+r_3\right)$ (B) $\frac{1}{\sqrt{A_1}}+\frac{1}{\sqrt{A_2}}+\frac{1}{\sqrt{A_3}}=\frac{1}{\sqrt{A}}$

(C) $\frac{1}{\sqrt{A_1}}+\frac{1}{\sqrt{A_2}}+\frac{1}{\sqrt{A_3}}=\frac{s^2}{\sqrt{\pi}r_1r_2r_3}$ (D) $\sqrt{A_1}+\sqrt{A_2}+\sqrt{A_3}=\sqrt{\pi}\left(4R+r\right)$

Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. XXXXX.XX).

49. In a triangle ABC, the equation of the side BC is $2x - y = 3$ and its circumcentre and orthocentre are at (2, 4) and (1, 2) respectively. The value of '$\sqrt{61}\sin B \sin C$' is equal to.

50. The integral values of n, the value of $\lim\limits_{n\to-\infty}\left(\sqrt{n^2-n+1}-\left[\sqrt{n^2-n+1}\right]\right)$, where [.] → Greatest Integer Function.

51. The solution of the differential equation $\frac{d^3y}{dx^3}-8\frac{d^2y}{dx^2}=0$ satisfying $y(0)=\frac{1}{8}$, $y_1(0)=0$ and $y_2(0)=1$ is $y=\frac{e^{8x}-8x+7}{\lambda}$. The numerical value of $\frac{\lambda}{10}$ is.

52. If $\int_{-\sqrt{2}}^{\sqrt{2}}\frac{\left(2x^7+3x^6-10x^5-7x^3-12x^2+x+1\right)}{\left(x^2+2\right)}dx=\frac{\sqrt{2}}{20}\left(5\pi+k^2\right)$, then the value of $\frac{k}{5}$ must be.

53. Let $|\vec{a}|=1$, $|\vec{b}|=1$ and $|\vec{a}+\vec{b}|=\sqrt{3}$. If $\vec{c}$ be a vector such that $\vec{c}=\vec{a}+2\vec{b}-3\left(\vec{a}\times\vec{b}\right)$ and $p=\left|\left(\vec{a}\times\vec{b}\right)\times\vec{c}\right|$, then p^2 is equal to.

54. In a ΔABC, if $\begin{vmatrix}1 & a & b\\ 1 & c & a\\ 1 & b & c\end{vmatrix}=0$, then the value of $\left(\sin^2 A+\sin^2 B+\sin^2 C\right)$ equals_____.

Space for Rough work